

Cloud Computing TASK – 4

- 1. In AWS, create a new security group named InstaClone-SG and configure it to allow incoming HTTP (port 80) and HTTPS (port 443) traffic only from your own IP address.**

Ans :

- Log in to the AWS Management Console.
- Navigate to EC2.
- In the left navigation pane, click Security Groups.
- Click Create Security Group.
- Enter the following details:
- Security Group Name: InstaClone-SG
- Description: Security group for InstaClone web server
- VPC: Select your existing VPC (usually the default VPC).
- Under Inbound Rules, click Add Rule.
- Configure the first rule:
- Type: HTTP
- Protocol: TCP
- Port Range: 80
- Source: My IP (automatically fills in your current public IP address).
- Click Add Rule again.
- Configure the second rule:
- Type: HTTPS
- Protocol: TCP

- Port Range: 443
 - Source: My IP.
 - Leave the Outbound Rules as the default (Allow All Traffic), unless your assignment specifies otherwise.
 - Review the configuration.
 - Click Create Security Group.
 - Verify that the security group InstaClone-SG has been created successfully.
- 2. Set up a security group for a mock Zomato-style backend server that allows incoming MySQL (port 3306) connections only from a specific IP address and blocks all other inbound traffic.

Hint: Use the 'Custom TCP' rule and specify the allowed IP in CIDR format.**

Ans :

- Log in to the AWS Management Console.
- Navigate to EC2.
- In the left navigation pane, click Security Groups.
- Click Create Security Group.
- Enter the following details:
- Security Group Name: Zomato-Backend-SG
- Description: Security group for Zomato backend MySQL server
- VPC: Select your existing VPC.
- Under Inbound Rules, click Add Rule.
- Configure the rule:
- Type: Custom TCP
- Protocol: TCP
- Port Range: 3306

- Source: Enter the specific IP address in CIDR format (e.g., 203.0.113.25/32 or the IP of your application server).
- Do not add any other inbound rules. This ensures all other inbound traffic is blocked.
- Leave the Outbound Rules as the default (Allow All Traffic), unless your assignment specifies otherwise.
- Review the configuration.
 - Click Create Security Group.

3. Modify an existing security group to block all outbound traffic except for DNS (port 53) and HTTPS (port 443) to simulate a strict Paytm wallet server environment.

Ans :

- Log in to the AWS Management Console.
- Navigate to EC2.
- In the left navigation pane, click Security Groups.
- Select the existing security group you want to modify (e.g., Paytm-Wallet-SG).
- Open the Outbound Rules tab.
- Click Edit Outbound Rules.
- Delete the default rule:
- Type: All Traffic
- Destination: 0.0.0.0/0
- Click Add Rule and configure the first rule:
- Type: DNS (UDP)
- Protocol: UDP
- Port Range: 53
- Destination: 0.0.0.0/0
- Click Add Rule again and configure:

- Type: DNS (TCP)
- Protocol: TCP
- Port Range: 53
- Destination: 0.0.0.0/0
- Click Add Rule again and configure:
- Type: HTTPS
- Protocol: TCP
- Port Range: 443
- Destination: 0.0.0.0/0
- Click Save Rules.
- Verify that the outbound rules now include only:
- DNS (UDP 53)
- DNS (TCP 53)
- HTTPS (TCP 443)

**4. Compare security groups and network ACLs by listing two key differences in how they control traffic for a Flipkart-like cloud application.

Hint: Think about the level (instance vs subnet) and direction (inbound/outbound) each operates at.**

Ans :

Feature	Security Groups	Network ACLs (NACLs)
1. Level of Control	Security Groups work at the EC2 instance level. They control traffic	Network ACLs work at the subnet level. They control traffic

Feature	Security Groups	Network ACLs (NACLs)
	coming to and leaving specific servers (for example, Flipkart web server instances).	entering and leaving an entire subnet containing multiple resources.
	Security Groups are stateful. If inbound traffic is allowed, the return outbound traffic is automatically allowed, even if no outbound rule exists.	Network ACLs are stateless. Inbound and outbound traffic must be allowed separately using separate rules.

2. Traffic Direction Handling